$$\underline{M_X(t) = E(e^{tX})} = \begin{cases} \int_{-\infty}^{\infty} e^{tx} f(x)\,dx & \text{if } X \text{ is continuous} \\ \sum_{x_i} e^{tx_i} p_X(x_i) & \text{if } X \text{ is discrete RV} \end{cases}$$

$$M_X(t) = \underline{E(e^{tX})}$$

$$= E\left(1 + tX + \frac{t^2X^2}{2!} + \frac{t^3X^3}{3!} + \cdots\right)$$

$$M_X(t) = E(1) + t\,E(X) + \frac{t^2}{2!}E(X^2) + \frac{t^3}{3!}E(X^3) + \cdots$$

$$M_X(t) = 1 + t\,E(X) + \frac{t^2}{2!}E(X^2) + \frac{t^3}{3!}E(X^3) + \cdots$$

$$\Rightarrow M_X'(t) = E(X) + t\,E(X^2) + \frac{t^2}{2!}E(X^3) + \cdots$$

$$M_X''(t) = E(X^2) + t\,E(X^3) + \frac{t^2}{2!}E(X^4) + \cdots$$

$$M_X'''(t) = E(X^3) + t\,E(X^4) + \frac{t^2}{2!}E(X^5) + \cdots$$

$$M_X(0) = 1$$

$$M_X'(0) = E(X)$$

$$M_X''(0) = E(X^2)$$

$$M_X'''(0) = E(X^3), \ldots\ldots, \qquad M_X^{(r)}(0) = E(X^r)$$